



# EdTech stack audit: compare Canvas, Blackboard, Moodle, and D2L Brightspace across LTI 1.3 standards, AI grading, and API extensibility

Computer

AI grading & assessment matrix: evaluate Gradescope, CoGrader,, and 1EdTech QTI 3.0 platforms for automated rubric scoring and LMS sync

Computer

How should work completion tools handle source grounding

What does a source-aware grading workflow look like

How can QTI 3 improve interactive assessment design

also more studying features and ai tools and learning tools

Semester can differentiate most by making learning tools **source-grounded, feedback-rich, accessible, and recoverable**—rather than simply adding an AI chatbot to an LMS. The native LMS should let students see what an assignment requires, what source supports each claim or study aid, how work maps to the rubric, and how every grade was reached.

QTI 3 is especially valuable because it supports portable assessment items, tests, results, interaction types, scoring rules, and accessibility metadata; accessibility capabilities formerly associated with APIP were integrated into QTI 3. [<sup>1599</sup>][1609]

## LMS and assessment landscape

Public documentation varies by product and deployment, so this is an architecture comparison—not a procurement conclusion. Validate versions, licenses, APIs, AI terms, hosting, and security controls with each vendor.

| Dimension           | Canvas                                                         | Blackboard Learn                                                                                    | Moodle                                                                 | D2L Brightspace                                                         | Semester target                                                                                             |
|---------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| LTI 1.3 / Advantage | Strong ecosystem support; validate enabled services per tenant | LTI 1.3/LTI Advantage availability documented for supported Learn SaaS and eligible hosted versions | Strong plugin and standards ecosystem; configuration/version dependent | Strong enterprise interoperability posture; verify tenant configuration | Native LTI 1.3 platform/tool roles; NRPS, Deep Linking, AGS; every grade write is previewable and auditable |
| Gradebook exchange  | LTI AGS and platform APIs; institution configuration applies   | LTI-grade integrations and platform API capabilities; validate deployment                           | Gradebook plus web-service/plugin ecosystem                            | Gradebook and APIs; validate entitlements and configuration             | Native grade ledger plus controlled LTI AGS/SIS sync, reconciliation, retry, rollback/exception queue       |

| Dimension              | Canvas                                                                    | Blackboard Learn                                                  | Moodle                                                                    | D2L Brightspace                                        | Semester target                                                                                                                     |
|------------------------|---------------------------------------------------------------------------|-------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| API extensibility      | Mature REST ecosystem and developer tooling                               | REST APIs and integration framework; deployment/version dependent | Open-source core plus web services/plugins                                | Enterprise APIs and partner ecosystem                  | Versioned REST, webhooks, OAuth/OIDC, SCIM, export APIs, documented data contracts, synthetic sandbox                               |
| AI-assisted grading    | Vendor tools and third-party integrations vary by institution and license | Varies by deployment and integrated toolset                       | Plugins/third-party tools vary; institution selects governance            | Vendor/partner capabilities vary by license and policy | AI suggests evidence, rubric alignment, and feedback drafts; authorized human owns every consequential grade                        |
| Assessment portability | Imports/integrations vary                                                 | Standards and vendor import tools vary                            | Strong community formats/plugins, but portability depends on content type | Enterprise content/integration tooling                 | QTI 3 import/export, assessment versioning, item-bank provenance, accessibility metadata, results portability                       |
| Governance emphasis    | Institution-configured roles, policies, integrations                      | Institution-configured roles and policies                         | Institution bears substantial implementation/governance responsibility    | Institution-configured enterprise controls             | Source/Scope/Status UX, privacy-first logs, course AI policy, accessibility gates, grade-change ledger, no hidden student profiling |

Blackboard documentation describes LTI 1.3/LTI Advantage tool registration for supported SaaS and certain hosted Learn deployments. Across all major LMS platforms, implementation capability depends on version, tenant licensing, and institution configuration—not merely vendor marketing. [^1612]

### Source-grounded work completion

A work-completion tool should make the chain from **assignment** → **approved source** → **student work** → **rubric** → **feedback** → **grade** visible and reviewable.

### Assignment evidence graph

Assignment

- Instructor-authored prompt
- Rubric version
- Course outcomes
- Required/approved sources
- Course policy, including AI rules
- Student task plan
- Student drafts and revisions
- Citations, notes, and evidence cards
- Submission version and receipt
- Feedback items

- └ Grade calculation
- └ Grade-change/review history

Every object needs metadata:

Source:

Instructor-created / institution-verified / library / student-provided /  
AI-generated / imported / external source

Authority:

Official course requirement / learning aid / personal draft / estimate

Status:

Draft / approved / current / superseded / submitted / graded / released /  
changed / needs review

Version:

Who changed it, when, why, and what version it replaced

## Work-completion workflow

Instructor publishes assignment

→ Semester stores prompt, rubric, source list, outcome map, policy, due date,  
late rules, and accessible-format requirements

Student opens assignment

→ sees plain-language task summary, source list, rubric checklist,  
milestone suggestions, and permitted AI uses

Student creates work

→ uses notes/source cards/citations linked to original materials  
→ can create a private AI study or drafting session under course policy  
→ sees where an AI statement, citation, or disclosure may be required

Student submits

→ receives versioned receipt, timestamp, file hash, and recovery route

Authorized grader reviews

→ scores criterion-by-criterion from rubric  
→ writes feedback tied to student evidence and course criteria  
→ uses AI only to assist, never to silently determine final grade

Grade releases

→ student sees score, rubric, feedback source, calculation, next action,  
review/appeal route, and whether it is official or an in-progress estimate

## Source-grounding rules

Do not let AI cite sources it did not retrieve or cannot identify.  
Do not treat generated text as an authoritative course source.  
Show page, section, timestamp, or excerpt anchors when available.  
Preserve source version and retrieval date.  
Distinguish instructor-approved sources from general-web material.  
Respect source access and copyright rights.

Exclude other students' work from retrieval by default.  
Do not put private student drafts into shared course indexes.  
Require user review before adding citations or quotations to a submission.

## Source cards

A source card should be available everywhere—in study tools, writing tools, grading feedback, and AI answers.

Title:  
Week 4 Lecture: Regression Basics

Type:  
Instructor-approved course material

Owner:  
Professor Patel

Location:  
Slide deck, pages 14-21

Course access:  
ECON 210, Fall 2026

Status:  
Current; updated September 19, 2026

How it is used:  
Referenced by Assignment 2 and Practice Set B

Open source:  
[Open original material]

## Source-aware grading workflow

A source-aware workflow means graders can substantiate feedback without pretending the system can objectively infer learning from a file alone.

## Faculty grading experience

Open submission

- view assignment prompt/rubric version alongside work
- see relevant course-source references and student-provided citations
- select criterion
- enter score and evidence-based comment
- optionally use AI to suggest possible rubric evidence or draft feedback
- review/edit every AI suggestion
- flag missing source, inaccessible file, policy question, or suspected integrity issue
- submit draft grade
- calibrate/moderate if configured
- release grade

## Feedback format

### Criterion:

Evidence and reasoning – 8/10

### Rubric standard:

Uses credible evidence and explains its relevance.

### Observed in your submission:

Your second paragraph uses the course reading, but the connection between the quotation and your claim needs explanation.

### Relevant source:

Week 4 Reading, page 17

### Suggested next action:

Add one or two sentences explaining how the cited finding supports your claim.

### Feedback status:

Instructor-authored and released

## Grade ledger

### Grade event:

Submission received

→ rubric scored

→ feedback added

→ grader review

→ moderation

→ released

→ student question/request

→ grade correction, if approved

→ external LMS/SIS sync, if enabled

## Each change records:

Actor

Role

Timestamp

Reason

Prior value

New value

Rubric/version

Approval where required

Sync status

Audit correlation ID

## AI guardrails for grading

| Activity        | AI role                   | Human control               |
|-----------------|---------------------------|-----------------------------|
| Rubric creation | Suggest initial structure | Instructor approves version |

| Activity              | AI role                                                           | Human control                                                  |
|-----------------------|-------------------------------------------------------------------|----------------------------------------------------------------|
| Question generation   | Draft from approved outcomes/materials                            | Faculty reviews for validity, bias, and accessibility          |
| Practice feedback     | Explain, hint, generate retrieval practice                        | Clearly formative; no official score                           |
| Evidence spotting     | Highlight possible text/response sections tied to rubric criteria | Grader validates relevance                                     |
| Feedback drafting     | Draft a comment based on grader-selected criterion/evidence       | Grader edits and releases                                      |
| Deterministic scoring | Calculate points from instructor-entered rubric values            | System calculation is inspectable                              |
| Integrity signals     | Flag issue for review                                             | No automatic accusation, grade penalty, or disciplinary action |
| Final grade           | Never AI-only                                                     | Authorized human is accountable                                |

Third-party products illustrate why the human-review boundary matters. Gradescope describes AI-assisted grading and answer grouping to accelerate grading of fixed-template work, while CoGrader markets rubric-aligned essay scoring and feedback with LMS integrations. These can speed workflows, but Semester should preserve faculty review, audit history, source traceability, and appeal paths before a grade becomes official. <sup>[1615]</sup><sup>[1611]</sup>

### QTI 3 interactive assessment design

QTI 3 should be more than an import/export checkbox. Use it as the content model for accessible, reusable, versioned assessment objects.

### QTI 3 design principles

One assessment object = item content + response rules + scoring logic + metadata + accessibility alternatives + outcome mapping + version history.

Do not bury accessibility in a separate accommodation afterthought.

Do not create interactions that work only by drag, hover, color, mouse, or a single sensory mode.

Do not make a question portable only in appearance—test import, delivery, response processing, scoring, and accessible alternatives.

### Interactive question library

| Assessment pattern  | Learning purpose                           | Accessible equivalent                                               |
|---------------------|--------------------------------------------|---------------------------------------------------------------------|
| Multiple response   | Distinguish several correct concepts       | Keyboard-operable checkbox group with clear feedback                |
| Matching            | Connect concepts, definitions, or examples | Select-based pairing alternative; no drag-only requirement          |
| Ordering            | Sequence process, logic, or chronology     | Number/select order interface with keyboard controls                |
| Numeric/formula     | Practice quantitative reasoning            | Accessible math input, unit checks, visible notation guidance       |
| Hotspot/image label | Identify visual features                   | Textual description, labeled list, or selectable-region alternative |

| Assessment pattern        | Learning purpose                             | Accessible equivalent                                                            |
|---------------------------|----------------------------------------------|----------------------------------------------------------------------------------|
| Simulation/case           | Apply decisions to realistic context         | Keyboard-operable scenario controls and text equivalent                          |
| Graph/data interpretation | Read data and make evidence-based conclusion | Data table, text summary, labeled axes, accessible formula/values                |
| Audio/video response      | Demonstrate oral communication or analysis   | Captions/transcript, alternative response format where policy permits            |
| Coding task               | Apply programming concepts                   | Sandboxed editor, keyboard workflow, deterministic tests, accessible output logs |
| Short answer/essay        | Explain reasoning or synthesize evidence     | Rubric, word limit, format guidance, accessible editor and autosave              |
| Peer review               | Practice evaluative judgment                 | Structured rubric; anonymity and accessibility controls                          |
| Portfolio                 | Demonstrate longitudinal work                | Artifact alternatives, ownership/consent, rubric and reviewer workflow           |

QTI 3 incorporates accessibility and internationalization features, and 1EdTech recommends adopting QTI 3 rather than APIP for new implementations because APIP accommodations were integrated into QTI 3. <sup>[1609]</sup>[1600]

## Interactive assessment authoring flow

```

Learning outcome selected
→ assessment blueprint created
→ item type selected
→ author writes prompt and response/scoring rules
→ author adds accessibility alternative and media description
→ author assigns metadata: topic, difficulty, outcome, cognitive level,
  source, rights, language, review date
→ author previews student, keyboard, and screen-reader modes
→ reviewer validates content, scoring, and accessibility
→ item version is approved and added to item bank
→ assessment delivery uses approved version
→ item analytics inform future revision

```

## Item analytics

Use analytics to improve questions—not to profile students.

```

Difficulty
Discrimination
Distractor performance
Time-to-complete distribution
Technical error rate
Accessibility issue reports
Question skip/abandon rate
Rubric consistency
Outcome alignment
Version comparison

```

Suppress small groups and do not use item interaction behavior to create individual student-risk labels.

## More study and AI learning tools

### Learning modes

| Tool                         | What it does                                                                    | Source and safety control                                            |
|------------------------------|---------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Teach-back coach             | Student explains a concept; system identifies gaps and asks follow-up questions | Uses authorized material; says “review this,” not “you failed”       |
| Retrieval practice generator | Makes quizzes/flashcards from selected course sources                           | Shows original source anchors and allows correction                  |
| Spaced-repetition planner    | Schedules review based on student-selected confidence and deadlines             | Student controls cadence; no pressure streaks                        |
| Worked-example tutor         | Walks through a representative solution step by step                            | Labels example vs graded work; follows course policy                 |
| Error-analysis coach         | Helps student categorize why a practice answer was wrong                        | No ability or intelligence labels                                    |
| Concept-map builder          | Connects terms, claims, evidence, formulas, and outcomes                        | Editable graph; source card on every generated connection            |
| Reading companion            | Breaks long reading into sections, vocabulary, questions, and notes             | Preserves citations/page anchors; supports text-to-speech            |
| Lecture companion            | Syncs slides, notes, transcript, timestamps, and study prompts                  | Requires authorized recording/transcript; respects instructor policy |
| Problem-set planner          | Breaks a problem set into effort estimates and support checkpoints              | No solution delivery when course policy restricts it                 |
| Exam readiness planner       | Builds a study schedule from scope, date, and student availability              | Clearly labels estimates; supports accessibility/quiet hours         |
| Office-hours agenda builder  | Creates a concise question list from student confusion points                   | Student chooses what to share                                        |
| Feedback-to-revision coach   | Turns released feedback into a revision checklist                               | Uses only student’s released feedback and selected sources           |
| Citation coach               | Explains attribution and source quality; builds a bibliography draft            | Student verifies all citations before use                            |
| Accessibility transformation | Read aloud, captions, plain-language restatement, format checks                 | Clearly identifies transformations; preserves original access        |
| Study-group kit              | Shared agenda, roles, practice prompts, and resources                           | Consent-based sharing; no grade/analytics exposure                   |
| Lab/studio notebook          | Time-stamped notes, observations, methods, data links, and reflection           | Permissions, export, and integrity controls                          |
| Formula/reference builder    | Student-authored and instructor-approved study sheets                           | Course-policy and assessment-permission label                        |
| Metacognition journal        | Records goals, confidence, strategy, and reflection                             | Private by default; no behavioral scoring                            |

## Advanced learning features

Prerequisite refresher paths:

Short instructor-approved bridge modules before difficult topics.

Misconception library:

Faculty-curated examples of common misunderstandings with explanations and source-linked corrections.

Adaptive practice:

Choose the next practice item using transparent rules:

topic coverage, student-selected confidence, prior practice response, and upcoming assessment scope—not opaque risk prediction.

Learning-outcome navigator:

Show which modules, assignments, practice items, and feedback connect to each stated outcome.

Feedback digest:

Summarize released feedback across assignments into student-controlled patterns and revision opportunities.

Study workload balancer:

Suggest a realistic plan across courses based on deadlines and student-declared availability, with all assumptions visible.

Academic integrity rehearsal:

Practice citation, paraphrase, collaboration, AI disclosure, and permitted-tool decisions without punitive grading.

Exam wrapper:

Before an exam, set strategy; after release, reflect on preparation, question types, and next-step support without labeling capability.

## Recommended priorities

Build these next, in order:

1. **Source-grounded Work Completion:** assignment breakdown, rubric checklist, source cards, draft/revision history, submission receipt, and recovery center.
2. **Study Studio:** teach-back, retrieval practice, spaced repetition, reading/lecture companion, feedback-to-revision, and accessibility tools.
3. **Source-aware Gradebook:** structured rubrics, criterion-level feedback, audit ledger, release controls, student grade estimate with limits, and regrade workflow.
4. **QTI 3 assessment core:** item bank, import/export, accessible interaction patterns, versioned scoring, delivery, and item analytics.
5. **Human-governed AI assistance:** course policy engine, grounded generation, feedback drafting, evidence spotting, integrity review—not AI-only grades.
6. **Advanced practice and outcomes:** transparent adaptive practice, misconception library, learning-outcome navigator, and portable evidence/credentials.

The best benchmark for Semester is not whether it can generate an essay or auto-score a quiz. It is whether every student can reliably answer: **What do I need to do, what sources support it, how will it**

be evaluated, what feedback did I receive, and what should I do next?

✱

1. <https://www.1edtech.org/standards/qti/index>
2. <https://cograder.com/schoology-extension/>
3. <https://www.qflowlearn.com/standards/qti-3/>
4. <https://edfast.ai/manuals/edfast-manual-lti.pdf>
5. <https://www.1edtech.org/standards/qti/accessibility>
6. <https://www.1edtech.org/blog/finally-final-the-qti-30-release>
7. <https://www.imsglobal.org/spec/qti/v3p0/impl>
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11. <https://cograder.com/content/canvas-ai-grading/>
12. <https://success.vitalsource.com/hc/en-us/articles/32942885817751-Blackboard-LTI-1-3-Tool-Setup-and-Link-Placement>
13. <https://cograder.com/content/google-classroom-ai-grading/>
14. <https://cograder.com/content/guide-to-standards-based-grading-ai-for-standards-based-rubrics/>
15. <https://guides.gradescope.com/hc/en-us/articles/24838908062093-AI-assisted-grading-and-answer-groups>